CO2020 : Computer-Aided Numerical Methods II

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PDEs Homework 2

Due Date: 23 March 2026, Monday, 11 pm

Question 1: Modify your code that solves the 2D steady heat conduction equation with constant diffusivity on a stretched grid, x_s. For this, you need to modify the coefficient matrix entries with appropriate metric terms, i.e. some combination of x'_u and x''_u. Here, x_u is the uniformly spaced mesh and 'prime' denotes differentiation with respect to x_s. Use the analytic stretching transformation

$$x_s = L_x \frac{(1+\beta)\left[(\beta+1)/(\beta-1)\right]^{(2x_u/L_x-1)} - \beta + 1}{2\left[1 + \left[(\beta+1)/(\beta-1)\right]^{(2x_u/L_x-1)}\right]}$$

that concentrates points close to the two ends. Here, $h = L_x/(nx-1)$ and $x_u = ih$ for $i = 0, 1, 2, ...nx-1$. Use the same transformation and same value of β for x and y. Visualize the grids (or grid spacings) for different values of β. Run your code as outlined in the previous homework and numerically determine the order of accuracy for two different stretching parameter values, e.g. $\beta = 1.25$ and $\beta = 1.1$.

Question 2: The 1D heat conduction equation over the range $x \in [0, 1]$ with the diffusivity $\kappa = 1$ and boundary values $T(0) = -1$ and $T(1) = 1$ has the exact solution $T(x,t) = \mathrm{erf}((x-0.5)/2\sqrt{t})$ for small times. "Small" here is with respect to the boundaries, i.e. the time until which the exact solution at the boundaries is not significantly different from the imposed conditions ($T(0) = -1$ and $T(1) = 1$).

A skeleton of a code to solve this equation numerically is provided to you. The domain is discretized into nx grid points, with $x[0] = 0$ and $x[nx-1] = 1$. We have discussed how to use the 2^{nd}-ordered centered-difference approximation for the spatial derivative and the first-order forward Euler time stepping for the temporal derivative in class.

(a) von-Neumann stability analysis for this problem with forward Euler time-stepping and second-order central differences for the spatial derivative gives the dispersion relation (i.e. relation between the growrh rate

σ and the wavenumber k) as

$$e^{\sigma \Delta t} = 1 - 2r(1 - \cos k\Delta x).$$

where $r = \kappa\,(\Delta t)\,/\,(\Delta x)^2$ is a non-dimensional form of the diffusivity. The wavenumbers vary from $k\Delta x \in [0, 2\pi)$. Plot the growth rate as a function of wavenumber for four values of $r = 0.4$, 0.5, 0.6, 0.7 on the same figure. Mark the line $\sigma = 0$ which demarcates the stable region from the unstable region. Comment on which modes (wavenumbers) are most unstable when the stability constraint is violated.

(b) Implement the method starting from the given skeleton code. Test it for different values of nx and Δt and satisfy yourself that your code is stable when $r < 1/2$ but the solution shows grid-to-grid oscillations (and eventually blows up) when executed with $r > 1/2$.

(c) Verify your implementation by numerically reproducing the expected order of accuracy. For this purpose, start your simulation from the exact solution at $t = 0.001$. Modify the code to write out the L_2 norm of the error between the exact solution and the numerical solution at the final time, t_{final}. Determine convergence behaviour (i.e. plot L_2 norm of the error between the exact and numerical solution for different values of nx) and explain your observations. You can use $t_{st} = 0.001$ and $t_{end} = 0.002$ with a time step $\Delta t = 1.25 \times 10^{-7}$. Your figure and discussion should demonstrate the following:

  (i) nx should range from around 10 to around 1000 in roughly equal increments.

  (ii) starting from small values of nx, the error reduces at 2^{nd}-order as nx is increased. On a log-log plot, your error values should be parallel to a straight line with slope of -2.

  (iii) the error starts to saturate (becomes constant or grows with nx) beyond a certain nx.

General instructions:

1. For plotting figures, use Matlab or Python any other postprocessing software of your choice. Ensure that the font size of the legend and labels is large enough to be easily visible. Ensure that the line thickness is appropriately large. Export each image as a png or eps file and include them in a pdf document.

2. Submit yor work as one pdf document with all handwritten work scanned to pdf and merged with the figures, comments, etc. Zip all your codes and submit them separately with your pdf on google classroom.